Computational Methods and Machine Learning for Finance – Project

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1 Introduction

This project aims to derive reasonable no-arbitrage prices for a set of simple derivatives written on an asset exhibiting "non-standard" dynamics. The assignment is divided into three main sections.

Section 1 focuses on the theoretical analysis of the given Stochastic Differential Equation (SDE) and the empirical assessment of the underlying time series characteristics. The SDE under consideration features a geometric drift component and a level-dependent diffusion term proportional to the square root of the asset price, a specification that introduces interesting challenges both from a theoretical and a computational standpoint. The preliminary analysis includes an evaluation of the Gaussian behavior of the model shocks via histogram and QQ-plot inspection, followed by the estimation of the optimal parameters via Maximum Likelihood Estimation (MLE).

Section 2 is dedicated to deriving inception no-arbitrage prices for a European Call option using three distinct numerical approaches: Monte Carlo simulation, Lattice discretization (Binomial Tree), and a Finite Difference explicit scheme (PDE). Each method approaches the pricing problem from a different angle (simulation-based, tree-based, and PDE-based respectively) and their convergence toward consistent results serves as a mutual validation of the implemented methodologies.

Section 3 extends the valuation to more complex derivative structures. We first price an American Call option, and compute the associated early exercise premium. We then turn to an exotic derivative, specifically an Up-and-Out Barrier Call option, whose payoff is extinguished if the underlying asset breaches a predefined upper barrier level at any point during the option's life. The path-dependent nature of this contract requires careful treatment in the simulation framework.

2 Preliminary Analysis & Parameters Estimation

2.1 Market Variable Analysis

2.1.1 Analysis of the given SDE

The time series $\{X_t\}_{t=0}^{N}$ is modeled through the following SDE:

$$dX_t = \kappa X_t dt + \sigma \sqrt{X_t}\, dW_t^Q \tag{1}$$

whereby:

- $\{W_t^Q\}_{t\geq 0}$ is a standard Brownian motion under a risk-neutral probability measure,
- κ is a parameter governing the drift of the process.
- σ is a parameter governing the volatility of the process.

The drift component κX_t implies that the expected change in the asset price is proportional to its current level, a feature commonly observed in financial assets such as equities or commodities. The diffusion term $\sigma\sqrt{X_t}$ introduces level-dependent volatility, meaning that the magnitude of random fluctuations increases with the asset price.

2.1.2 Visual Inspection

The initial analysis of the provided time series reveals the trajectory of the asset over the observation period.

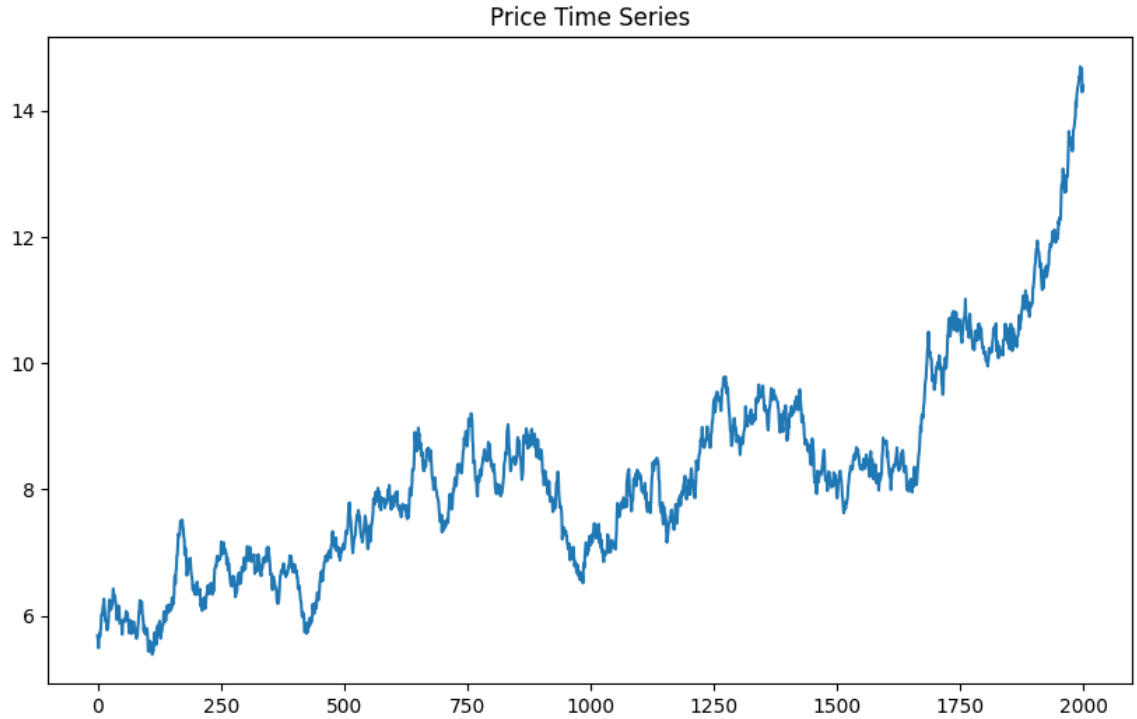

Figure 1: Price Time Series

The price time series shown in Figure 1 provides several insights into the behaviour of the underlying asset. First, the series remains strictly positive throughout the observation period, which is consistent with the characteristics of most financial assets such as equities or commodities that cannot take negative values. The trajectory of the price also shows a gradual upward trend over time, suggesting the presence of a positive drift component in the dynamics of the process.

In addition, the magnitude of fluctuations appears to increase as the asset price rises. In the earlier part of the sample, when the asset price is relatively low, the movements around the trend are comparatively small. As the price grows, the variability of the series becomes more pronounced, with larger oscillations around the trend. This behaviour suggests that the volatility of the asset is related to its level.

Such empirical behaviour provides intuitive support for the stochastic specification proposed in Equation 1, where the diffusion term is proportional to the square root of the asset price. In this formulation, the magnitude of random shocks increases with the level of the process while remaining bounded in a way that prevents extreme explosive behaviour. The model therefore captures both the positivity of the asset price and the increasing variability observed in the data.

2.1.3 Interpretation and nature of the Risk Factor

This structure of the SDE in Equation 1 is reminiscent of the Cox-Ingersoll-Ross (CIR) model typically used for interest rates to prevent negative values. However, the drift term κX_t implies a geometric growth (or decay) proportional to the current level, typical of equity models like Black-Scholes.

Therefore, we can reasonably assume that X represents a stock price or a commodity index characterized by a "leverage effect" or specific volatility skew where volatility is strictly tied to the price level, ensuring the asset remains non-negative (an essential property for limited liability assets).

2.2 Euler Discretization and Normality Check

To validate the model and estimate parameters, we first discretize the continuous SDE using the Euler–Maruyama scheme. Conditional on the model holding true, the discrete process is:

$$X_{t+\Delta t} = X_t + \kappa X_t \Delta t + \sigma \sqrt{X_t} \sqrt{\Delta t}\, z_{t+1} \tag{2}$$

where $z_{t+1} \sim \mathcal{N}(0, 1)$.

To verify this assumption, we invert the relationship to isolate the shocks based on the observed data:

$$z_{t+1} = \frac{X_{t+\Delta t} - X_t - \kappa X_t \Delta t}{\sigma \sqrt{X_t \Delta t}} \tag{3}$$

In order to start computing the series of empirical shocks, we made educated guesses as for expected values for both κ and σ. Based on the literature, we chose the following values:

- Drift: $\kappa \approx 0,1$
- Volatility: $\sigma \approx 0,3$

2.2.1 Statistical Evidence

In Figure 2, we assessed the distribution of these empirical shocks against the theoretical standard normal distribution $\mathcal{N} \sim (0, 1)$.

First of all, when looking at the histogram, we can see that the empirical distribution is clearly bell-shaped, confirming the Gaussian nature of the shocks. However, it exhibits a higher variance than the Standard Normal $\mathcal{N}(0, 1)$, as can be seen on Figure 3. This is expected as we are using initial heuristic parameters rather than optimized MLE estimates. The QQ-Plot confirms this: the points follow a straight line (normality), but with a slope different from 1 (non-unit variance).

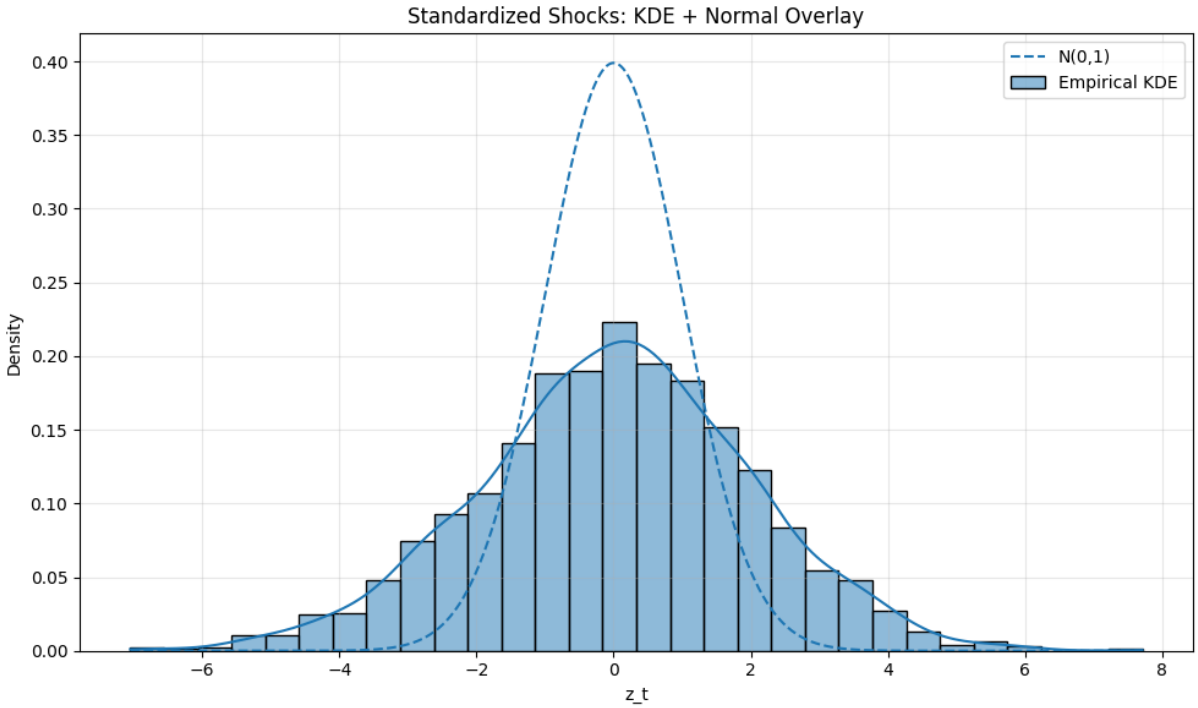

Figure 2: Empirical KDE

Then, the QQ plot compares the empirical quantiles against theoretical normal quantiles. The data points align closely with the 45-degree line, particularly in the central body of the distribution, with a little departure on the left tail.

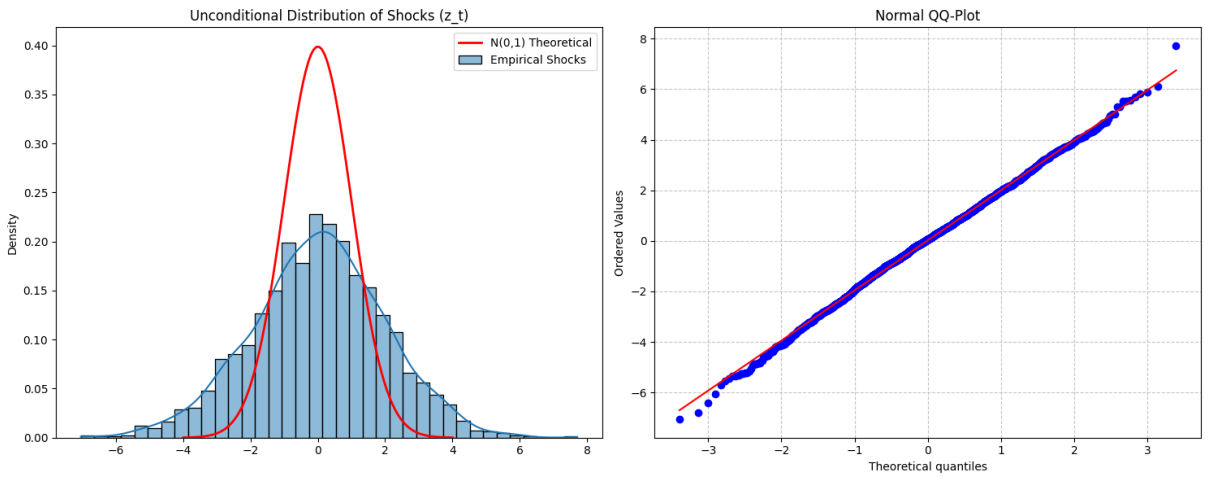

Figure 3: Unconditional Distribution of Shocks (Histogram and QQ-Plot)

Empirically, we found the following true characteristics of the shocks' distribution function:

- Mean: $\mu = 0,0207$ (Expected 0)
- Standard deviation: $\sigma = 1,9803$ (Expected 1)
- Distribution: Shocks $\sim \mathcal{N}(\mu, \sigma^2) \approx \mathcal{N}(0, 4)$

The empirical mean is close to the expected value of 0, whereas the empirical standard deviation is approximately equal to $1,98$ rather than the theoretical $1,0$. The scaling effect is entirely expected and mathematically consistent: it simply indicates that our initial educated guess for the volatility parameter ($\sigma = 0,3$) underestimates the true variance of the process.

This preliminary check successfully validates the assumption of normal transition densities, fully justifying the use of Maximum Likelihood Estimation (MLE) in the next step to calibrate the optimal parameters $\Theta = (\kappa, \sigma)$.

2.3 Maximum Likelihood Estimation (MLE)

2.3.1 Optimal Parameter

Relying on the normality of z_t, the transition density of $X_{t+\Delta t}$ given X_t is approximately normal with:

- Mean: $\mu_t = X_t + \kappa X_t \Delta t$
- Variance: $v_t = \sigma^2 X_t \Delta t$

The negative log-likelihood function was constructed as:

$$-\ln L(\theta) = \frac{1}{2} \sum_{t=0}^{N-1} \left[\ln(2\pi v_t) + \frac{(X_{t+1} - \mu_t)^2}{v_t} \right] \tag{4}$$

We minimize the previous function in order to determine the optimal parameter set: $\Theta = (\kappa, \sigma)$.

2.3.2 Estimated Parameters

The approximated values come directly from our code

- $\hat{\kappa}_{\text{MLE}} \approx 0,1323$
- $\hat{\sigma}_{\text{MLE}} \approx 0,5941$

These parameters suggest a positive drift component and a volatility scaling factor consistent with the observed path. As expected from the previous distribution of the residuals, we can clearly see that our educated guess for the volatility parameter σ was indeed underestimating the true variance of the process.

	κ	σ
Educated Guess (Literature)	0,1	0,3
MLE	0,1323	0,5941

Table 1: Summary Table of the MLE Parameter estimation

3 European Call Option Valuation

We proceed to price a plain vanilla European Call option with the following characteristics:

- Maturity (T): $0,6$ years
- Strike (K): $X(0) = x_N$ (At-the-money based on the last observed price)
- Risk-free rate (r): 2%

3.1 Monte Carlo Approach

The Monte Carlo method involves simulating a large number ($N_{\text{sim}} = 100'000$) of possible future paths of the asset price using the estimated parameters and the Euler discretization scheme. The price is estimated as the discounted expectation of the terminal payoff:

$$\pi_0^E \approx e^{-rT} \frac{1}{N_{\text{sim}}} \sum_{i=1}^{N_{\text{sim}}} \max\Big(X_T^{(i)} - K, 0\Big) \tag{5}$$

This method provides a robust estimate and allows for the calculation of a standard error to construct confidence intervals.

3.1.1 Justification of hyperparameters

For the Monte Carlo pricing, we selected $N_{sim} = 100'000$ simulations. This choice is justified by the following technical considerations.

First of all, we wanted to focus on statistical Precision. The standard error of a Monte Carlo estimator decreases at a rate of $1/\sqrt{N_{sim}}$. With $100'000$ paths, we reduce the sampling error significantly, ensuring a narrow 95% confidence interval for the option price.

Moreover, we also wanted to take into account for the convergence of non-standard dynamics. Given that our SDE's volatility depends on $\sqrt{X_t}$ (Equation 1), a high number of simulations is required to accurately capture the distribution of the terminal payoff, especially for out-of-the-money scenarios.

Finally, our model should be computationally efficient. The value we chose for N_{sim} represents an optimal trade-off; it provides high numerical stability while remaining computationally inexpensive in the Colab environment.

3.1.2 Euler-Maruyama Discretization

The risk factor dynamics, as defined in Equation 1, resemble a square-root diffusion process similar to the Cox-Ingersoll-Ross (CIR) model, characterized by a geometric drift κX_t instead of mean-reversion. While such processes admit a known transition density based on the non-central chi-squared distribution, we utilized an Euler-Maruyama discretization for our Monte Carlo simulations. This choice was dictated by practical flexibility and ease of implementation for path-dependent structures, ensuring that any discretization bias was mitigated by selecting an appropriately high frequency of time steps.

To simulate asset paths and extract empirical shocks, we employed the Euler-Maruyama scheme. Despite the non-standard nature of the SDE, this approach is justified for multiple reasons

First of all, we assess the small time step (Δt) of our model. Since we are dealing with daily observations ($dt = 1/250$), the discretization error is negligible. At this scale, the Euler scheme provides a robust approximation of the continuous-time process.

We also want to make sure that our approach is adaptable to state-dependent volatility. The Euler scheme is the standard choice for SDEs where the drift and diffusion coefficients depend on the level of the process X_t, allowing us to directly implement the $\sigma\sqrt{X_t}$ term.

Finally, we want to address the absence of mean-reversion within our given SDE. Our model $dX_t = \kappa X_t dt + \sigma\sqrt{X_t}dW_t$ lacks the mean-reverting term found in the classic CIR model. This implies a localized growth dynamic. The Euler scheme effectively captures this growth without the need for the complex transformations required by more advanced closed-form bond-pricing models.

3.1.3 Visualization

The following plot shows a sample of 50 asset paths under the previously mentioned conditions for our Monte Carlo simulations.

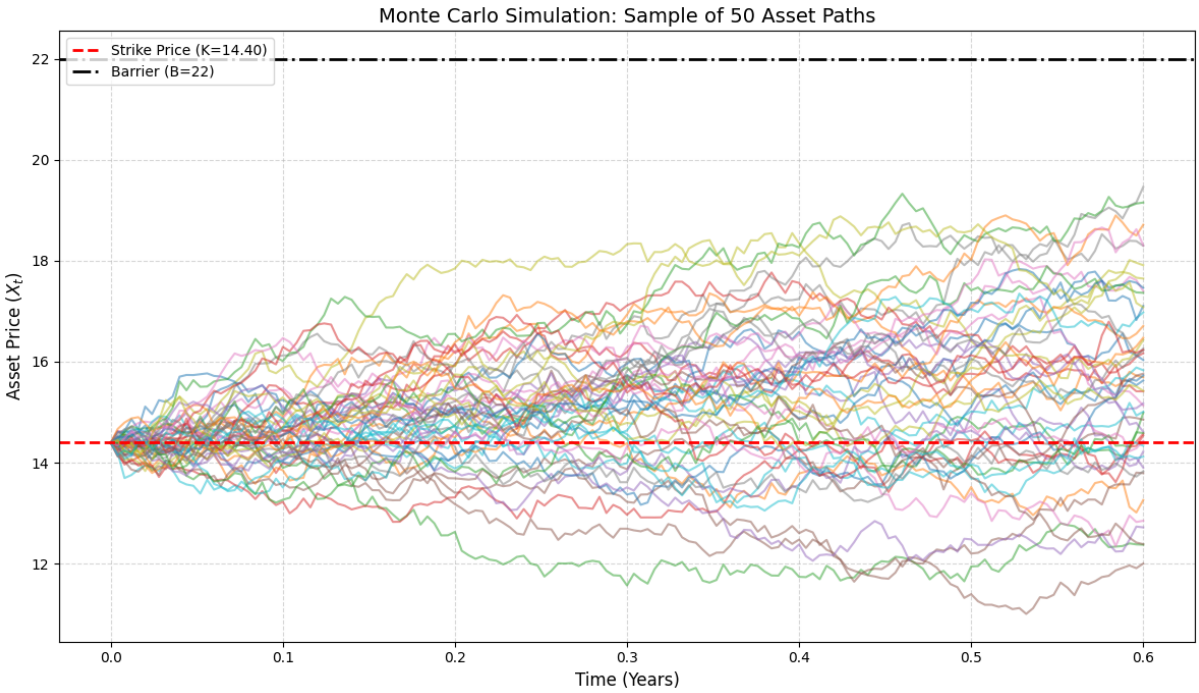

Figure 4: Visualization of 50 Price paths under our Monte Carlo Simulation

3.1.4 Result

The price is estimated as the discounted expectation of the terminal payoff:

$$\pi_0^E \approx e^{-rT} \frac{1}{N_{\text{sim}}} \sum_{i=1}^{N_{\text{sim}}} \max(X_T^{(i)} - K, 0) \tag{6}$$

The Monte Carlo approach provides us with a price equal to $1,4514$.

3.2 Lattice Methodology (Binomial Tree)

3.2.1 Binomial Tree parameters

We implemented a Binomial Lattice model to price the option. This requires discretizing the time interval $[0, T]$ into $N_{\text{steps}} = 300$. To adapt the tree to our specific process (where volatility depends on $\sqrt{X}$), we utilized a simplified approximation for the up (u) and down (d) movements:

$$u = \exp\left(\frac{\sigma\sqrt{\Delta t}}{\sqrt{X_0}}\right), \qquad d = \frac{1}{u} \tag{7}$$

The probability of an up-move (p) is derived to match the risk-neutral drift. The option price is obtained via backward induction, starting from the terminal payoff $\max(S_T - K, 0)$ and discounting back to $t = 0$.

3.2.2 Invertible Transformation Method

As discussed in class, we need to apply the invertible transformation method in order to guarantee for computational feasibility. Instead of discretizing $\sigma \cdot \sqrt{X_t}$, we use an inverse:

$$Y_t = f(x) = 2 \cdot \sqrt{x} \tag{8}$$

Then, we write:

$$f^{-1}(y) = \left(\frac{y}{2}\right)^2 = \frac{y^2}{4} \tag{9}$$

Finally, we apply Itô's Lemma on Y_t and find the following:

$$dY_t = \left(\frac{\kappa}{2}Y_t - \frac{\sigma^2}{2 \cdot Y_t}\right) \cdot dt + \sigma \cdot dW_t \tag{10}$$

3.2.3 Result

The Lattice method (Binomial Tree) provides us with a price equal to $1,4512$.

3.3 PDE-Scheme Pricing (Finite Difference)

The pricing problem can be formulated as a Partial Differential Equation (PDE) using the Feynman–Kac theorem. We solved this using an Explicit Finite Difference Scheme.

3.3.1 Grid construction

A grid was defined by:

- $N = 10'000$ time steps
- $M = 100$ price steps
- Extending the asset price domain to $S_{\max} = 3 \cdot X_0$ to mitigate boundary errors

We chose these values for both M and N as it is the convention to maintain a squared proportional relationship between both values.

3.3.2 Discretized PDE Coefficients

The continuous derivatives were replaced by finite difference approximations (gamma, theta).

$$\gamma_i = \frac{\sigma^2 S_i}{2(\Delta S)^2}\Delta t, \qquad \theta_i = \frac{\kappa S_i}{2\Delta S}\Delta t \tag{11}$$

3.3.3 Boundary Conditions

- $V(S, T) = \max(S - K, 0)$ (Terminal condition)
- $V(0, t) = 0$ (Call value at zero asset price)
- $V(S_{\max}, t) \approx S_{\max} - Ke^{-r(T-t)}$ (Linearity for deep ITM calls)

The explicit scheme iteratively computes the option value grid backwards in time. The final price is found by interpolating the grid values at the current spot price X_N.

3.4 Explicit Finite Difference Scheme

The PDE is solved using an explicit finite difference scheme on a grid defined by:

$$S_i = i\Delta S, \qquad t_n = n\Delta t \tag{12}$$

The derivatives are approximated as:

$$\frac{\partial V}{\partial t} \approx \frac{V_i^{n+1} - V_i^n}{\Delta t}, \qquad \frac{\partial V}{\partial S} \approx \frac{V_{i+1}^n - V_{i-1}^n}{2\Delta S}, \qquad \frac{\partial^2 V}{\partial S^2} \approx \frac{V_{i+1}^n - 2V_i^n + V_{i-1}^n}{(\Delta S)^2} \tag{13}$$

Substituting these approximations into the PDE yields the explicit update formula

$$V_i^{n+1} = (1 - 2\gamma_i - r\Delta t)V_i^n + (\gamma_i + \theta_i)V_{i+1}^n + (\gamma_i - \theta_i)V_{i-1}^n \tag{14}$$

with coefficients:

$$\gamma_i = \frac{\sigma^2 S_i}{2(\Delta S)^2}\Delta t, \qquad \theta_i = \frac{\kappa S_i}{2\Delta S}\Delta t \tag{15}$$

Boundary conditions are imposed as: $V(0,t) = 0$ and $V(S_{\max}, t) \approx S_{\max} - Ke^{-r(T-t)}$.

3.4.1 Result

The Finite Difference Methodology provides us with a price equal to $1,4536$.

3.5 Method Comparison and Convergence

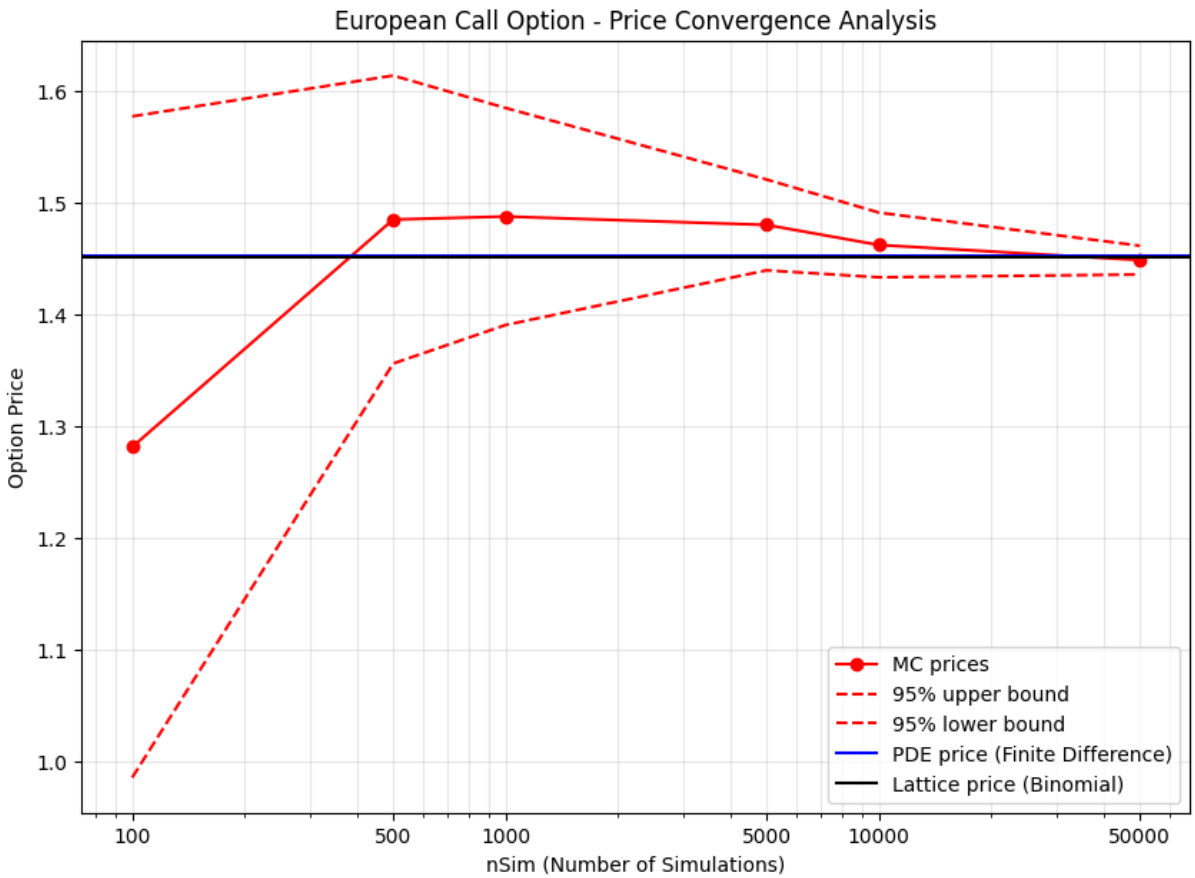

Figure 5: European Call Option – Price Convergence Analysis

To validate the robustness of our estimates, we analyzed the convergence of the Monte Carlo price as the number of simulations increases.

Figure 5 illustrates that the Monte Carlo estimate (red line) converges towards the values obtained by the deterministic methods (Lattice and PDE, represented by horizontal lines) as N_{sim} increases. The 95% confidence intervals narrow significantly, confirming the statistical consistency of the result.

	Estimated Price
Monte Carlo Simulations	1,4514
Lattice Methodology (Binomial Tree)	1,4512
Finite Difference Method	1,4536

Table 2: Summary Table of estimated prices for the plain vanilla European Call option

4 American Call Option & Barrier Option

4.1 American Call Option Valuation

The American Call option grants the holder the right to exercise the option at any point prior to maturity. We priced this instrument using the Lattice Method (Binomial Tree), as it naturally handles the early exercise feature.

4.1.1 Algorithm

At each node (i, j) of the tree, the option value is determined by maximizing the value of holding the option (Continuation Value) versus the value of exercising it immediately (Intrinsic Value):

$$V_{i,j} = \max\Big(e^{-r\Delta t}[pV_{i+1,j+1} + (1-p)V_{i+1,j}],\ S_{i,j} - K\Big) \tag{16}$$

4.1.2 Theoretical Consistency, Results and Visualization

To visualize the optimal exercise strategy, we plotted the decision boundary within the binomial tree.

- **Green Nodes**: Continuation is optimal.
- **Red Nodes**: Exercise is optimal.

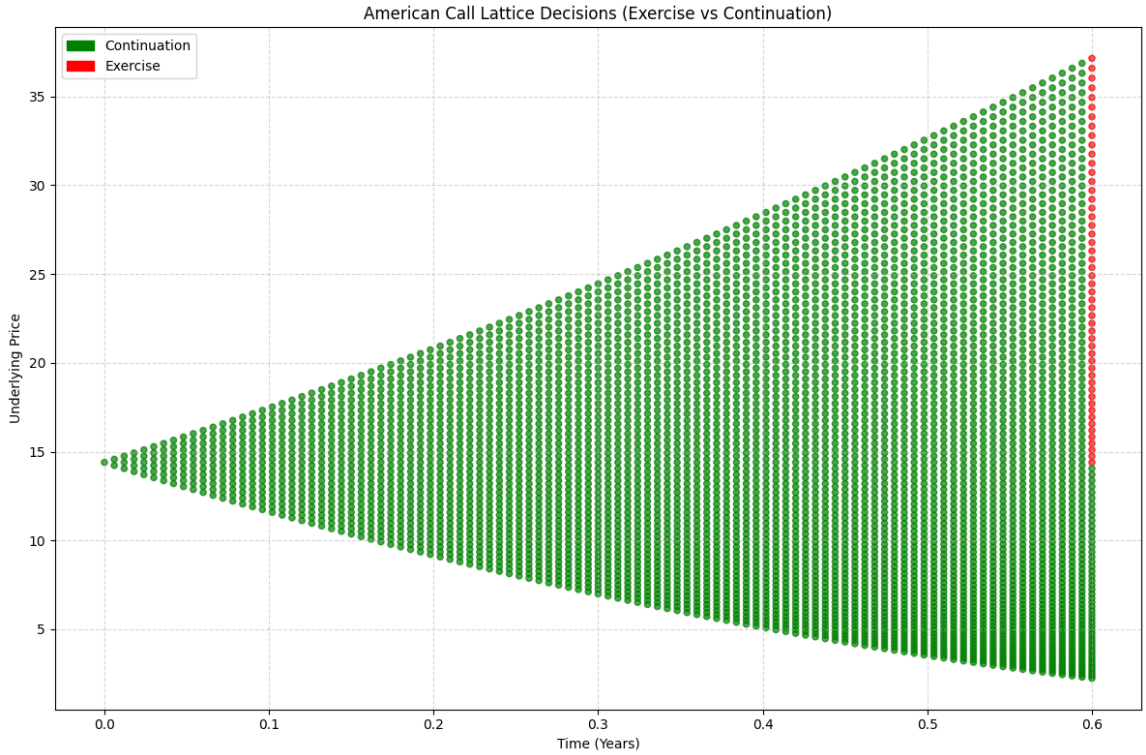

Figure 6: American Call Lattice Decisions (Exercise vs. Continuation)

Our numerical results indicate that the American Call price ($\approx 1,4512$) is equivalent to the price of the European Call ($\approx 1,4512$) using the Binomial Tree methodology, resulting in an Early Exercise Premium (EEP) of 0. This finding is perfectly consistent with Merton's fundamental theorem on rational exercise: for an American Call on a non-dividend-paying asset with a positive risk-free rate ($r > 0$), it is never optimal to exercise early. This is also visually observable on Figure 6, where the only red dots are at $0,6$ years, which corresponds to the

maturity of our financial products. In our case, we are never triggering the early exercise, we only exercise in some cases at maturity. This is overall consistent with an American Call Option with a positive drift.

The loss of insurance (time value) and the foregone interest on the strike price K outweigh any benefit from immediate exercise. The scatter plot (Figure 6) of the decision regions confirms that the Continuation region dominates the entire domain.

4.2 Up-and-Out Barrier Call Pricing

Finally, we priced an Up-and-Out Call Option. This is an exotic path-dependent derivative that becomes worthless if the asset price X_t touches or exceeds a barrier level $B = 22$ at any time $t \in [0, T]$.

4.2.1 Methodology and Risk Management

Due to the path-dependent nature of the barrier, we utilized the Monte Carlo method with $100'000$ simulated paths. This approach allows for a straightforward monitoring of the maximum price reached in each simulation:

$$\text{Payoff} = (X_T - K)^+ \cdot \mathbb{I}_{\{\max_{t\in[0,T]} X_t < B\}} \tag{17}$$

The computed price ($\approx 1,4438$) is lower than the vanilla European Call price ($\approx 1,4512$). This discount represents the "barrier risk", which is the probability that an otherwise profitable path hits the $B = 22$ threshold. This exercise demonstrates the versatility of simulation-based pricing for exotic derivatives where analytical or lattice-based adjustments for continuous monitoring would be significantly more complex.

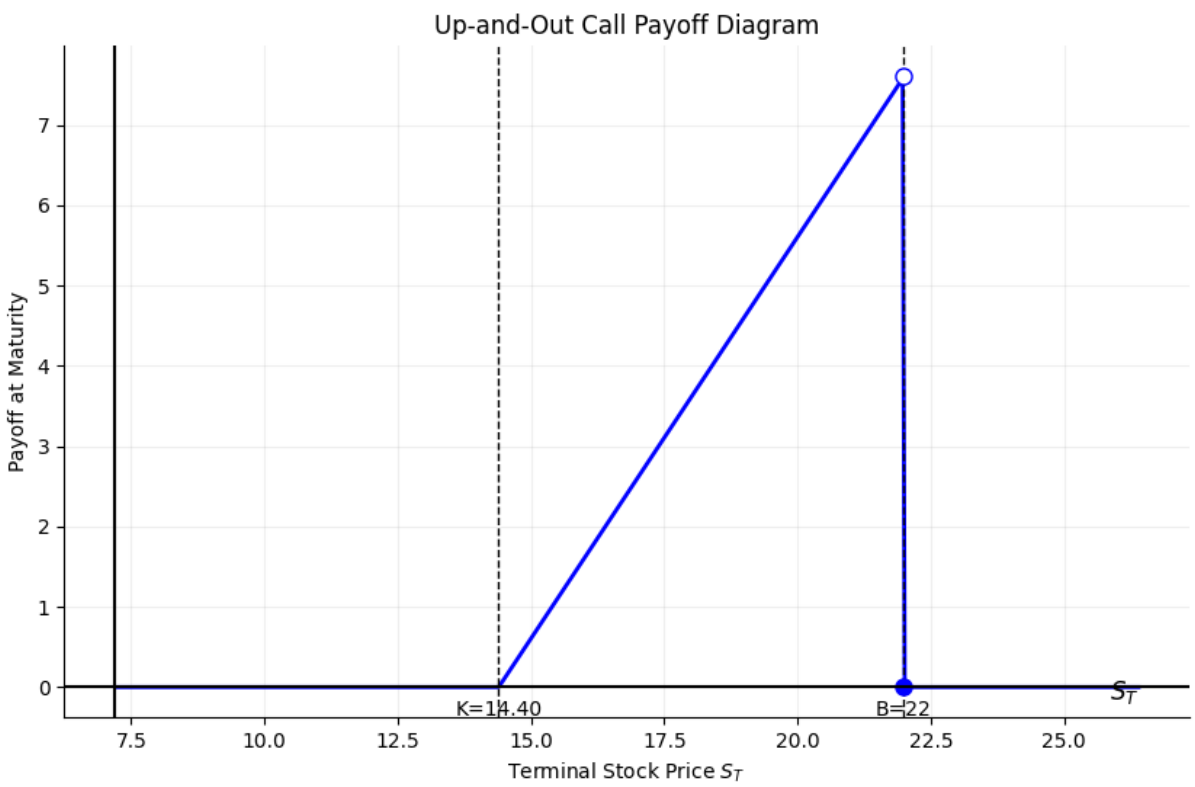

Figure 7: Up-and-Out Call Payoff Diagram

4.2.2 Analysis

The payoff diagram (Figure 7) illustrates the discontinuity at the barrier B. The price of this option is significantly lower than the standard European Call because the "Knock-Out" feature limits the upside potential. Since the initial price X_0 is relatively close to the barrier $B = 22$ (depending on the data provided), the probability of "knocking out" is non-negligible, drastically reducing the option's fair value.

5 Final Conclusions

In this project, we analyzed a financial asset following a stochastic process with level-dependent volatility and estimated its parameters using Maximum Likelihood Estimation. The calibrated parameters were then used to price several derivatives written on the underlying asset.

A European Call option was first valued using three distinct numerical approaches: Monte Carlo simulation, a Binomial Lattice method, and an explicit Finite Difference scheme. All three methodologies produced highly consistent results, with option prices converging around $1,45$. This strong agreement across simulation-based, tree-based, and PDE-based techniques provides validation of the numerical implementations and confirms the robustness of the pricing framework.

The analysis was then extended to more complex derivatives. An American Call option was priced using a lattice approach to account for the possibility of early exercise, while an Up-and-Out Barrier Call option was evaluated through Monte Carlo simulation due to its path-dependent nature. As expected, the barrier option exhibited a lower value than the plain vanilla European Call, reflecting the additional knock-out risk.

Overall, the project highlights how different computational techniques can be applied to solve derivative pricing problems under non-standard asset dynamics. The consistency of the results across methods demonstrates the reliability of the implemented models and illustrates the practical importance of numerical methods in modern quantitative finance.